# STUDENT PROJECT REPORT

# CANTEEN / FOOD ORDERING SYSTEM

Python Essentials – Evaluated Course Project

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| **Project Title** | Canteen / Food Ordering System |
| **Programming Language** | Python 3 |
| **Project Type** | Command-Line Application |
| **Academic Year** | 2026 |

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## 1. Introduction

The Canteen / Food Ordering System is a Python-based command-line application designed to simplify food ordering and billing. The program displays a menu, accepts food selections and quantities, calculates item totals, applies a discount according to the configured rules, and generates an order summary.

## 2. Problem Statement

Manual calculation of food orders can take time and may result in calculation errors. This project provides a simple computerized method for selecting food items, calculating quantities and prices, applying discounts, and producing a final bill.

## 3. Aim

To develop a Python command-line application that allows a customer to select food items, enter quantities, calculate the bill, apply a discount, and generate a clear order summary.

## 4. Objectives

• Display the available food menu with prices.
• Allow selection of one or more food items.
• Accept and validate item quantities.
• Calculate the total cost of each selected item.
• Calculate subtotal and discount.
• Generate a final order summary.
• Demonstrate fundamental Python programming concepts.

## 5. Scope

The project is suitable for a small canteen and academic demonstration. It focuses on order entry and billing through a terminal. Advanced features such as online payment, database storage, delivery tracking, and a graphical interface are outside the scope of the basic version.

## 6. Technologies Used

Programming Language: Python 3.x
Development Environment: Visual Studio Code or another Python editor
Execution Environment: Command-line terminal
External Libraries: None required for the basic version.

## 7. Main Features

• Food menu display
• Multiple-item ordering
• Quantity input
• Input validation
• Subtotal calculation
• Discount calculation

• Final bill calculation
• Order summary generation

## 8. Algorithm / Working

1. Start the program.
2. Display the food menu.
3. Ask the customer to enter an item number.
4. Validate the item selection.
5. Ask for the quantity and validate it.
6. Calculate price × quantity.
7. Add the item to the order.
8. Repeat until the customer selects Finish Order.
9. Calculate the subtotal.
10. Determine the applicable discount.
11. Calculate the final amount.
12. Display the order summary.
13. End the program.

## 9. Discount Logic

Below ■200: 0% discount
■200–■399: 5% discount
■400–■599: 10% discount
■600 and above: 15% discount

## 10. Python Concepts Used

Dictionary for menu data, lists for storing orders, tuples for grouped item information, while loops for repeated ordering, for loops for displaying items, if/elif/else for validation and discount logic, arithmetic operations for billing, and formatted strings for a readable receipt.

## 11. Project Structure

Recommended multi-file organization:
**main.py** – controls the application
**data.py** – stores menu data and prices
**customer.py** – handles customer information
**menu.py** – displays the menu
**order.py** – handles item selection and quantities
**validation.py** – validates input
**billing.py** – calculates subtotal, discount and final amount
**README.md** – setup and execution instructions

## 12. Testing

The project should be tested with valid item selections, invalid item numbers, zero or negative quantities, multiple food items, different subtotal ranges, and the finish-order option. Each test should produce the expected validation or billing result.

## 13. Sample Output

The terminal displays the canteen menu and asks for item numbers and quantities. After the customer finishes the order, the program displays each ordered item, quantity, price, subtotal, discount, and final amount. Screenshots of the student's actual working program can be inserted in this section.

## 14. Advantages

- Simple and user-friendly.
- Reduces manual billing calculations.
- Supports multiple food items.
- Applies discount automatically.
- Runs directly in a terminal.

## 15. Limitations

- The basic version does not permanently store orders.
- It does not include online payment.
- It is command-line based.
- Menu changes require editing the program data.

## 16. Future Scope

The project can be extended with database storage, customer accounts, order history, a graphical or web interface, online payments, inventory management, and stock notifications.

## 17. Conclusion

The Canteen / Food Ordering System demonstrates the practical use of Python for automating a simple billing workflow. It combines data structures, loops, conditional statements, validation, and arithmetic operations to create a functional command-line application.

## 18. References

Python Documentation: https://docs.python.org/3/
GitHub Documentation: https://docs.github.com/
Course-provided Python Essentials project instructions.

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Before submission, add screenshots of your own working program and the final GitHub repository URL if required by your course.